Computational Logic

Proofs in propositional logic - derived rules

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2026/27

Previously on LC...

We have define:

- We have defined the notions of deduction system, theorem and proof
- We have introduced the Gentzen system of natural deduction
- This system has no axioms and allows hypothesis to be introduced in the middle of the proof
- The system is defined a set of introduction/elimination rules for each connective

Some derived rules for *DN*

From the base rules we can obtain derived rules that correspond to theorems in the *DN* system.

Modus Tollens (in Latin, *way that denies*)

$$\frac{\phi \to \psi \qquad \neg\psi}{\neg\phi}\,\mathbf{MT}$$

Show that $\phi \to \psi, \neg\psi \vdash \neg\phi$:

1	$\phi \to \psi$	
2	$\neg\psi$	
3	$\quad\phi$	
4	$\quad\psi$	$\to$E, 1, 3
5	$\quad\mathbf{F}$	**F I**, 2, 4
6	$\neg\phi$	¬I, 3–5

Some rules derived from *DN*

Introduction of double negation $\dfrac{\phi}{\neg\neg\phi}\,\neg\neg\mathrm{I}$

(already shown...)

Some rules derived from *DN*

Reduction to absurdity

$$\dfrac{\begin{array}{c}[\neg\phi]\\ \vdots\\ \mathbf{F}\end{array}}{\phi}\,\mathsf{RA}$$

If we have a deduction of $\mathbf{F}$ assuming $\neg\phi$ we can have a deduction of $\neg\phi \to \mathbf{F}$. So just show $\neg\phi \to \mathbf{F} \vdash \phi$:

1	$\neg\phi \to \mathbf{F}$	
2	$\quad\neg\phi$	
3	$\quad\mathbf{F}$	$\to$E, 1, 2
4	$\neg\neg\phi$	$\neg$ **I**, 2–3
5	ϕ	$\neg$E, 4

Some derived rules for *DN*

Excluded middle (or third excluded) $\dfrac{}{\phi \vee \neg\phi}\,\mathsf{TE}$

1	$\neg(\phi \vee \neg\phi)$	
2	$\quad\phi$	
3	$\quad\phi \vee \neg\phi$	$\vee$I, 2
4	$\quad\mathbf{F}$	**F I**, 1, 3
5	$\neg\phi$	$\neg$I, 2–4
6	$\phi \vee \neg\phi$	$\vee$I, 5
7	$\mathbf{F}$	**F I**, 1, 5
8	$\phi \vee \neg\phi$	**RA**, 1–7

$\neg q \to \neg p \vdash p \to \neg\neg q$:

1	$\neg q \to \neg p$	
2	$\quad p$	
3	$\quad \neg\neg p$	$\neg\neg$I, 2
4	$\quad \neg\neg q$	**MT**, 1, 3
5	$p \to \neg\neg q$	$\to$I, 2–4

Deductive equivalence

Given the formulas ϕ and ψ, we say that ϕ and ψ are deductively equivalent if and only if $\phi \vdash \psi$ and $\psi \vdash \phi$. And we denote it by $\phi \dashv\vdash \psi$.

Contraposition: $\phi \to \psi \dashv\vdash \neg\psi \to \neg\phi$

1	$\phi \to \psi$	
2	$\quad \neg\psi$	
3	$\quad \neg\phi$	**MT**, 1, 2
4	$\neg\psi \to \neg\phi$	$\to$I, 2–3

1	$\neg\psi \to \neg\phi$	
2	$\quad \phi$	
3	$\quad\quad \neg\psi$	
4	$\quad\quad \neg\phi$	**F I**, 1, 3
5	$\quad\quad$ **F**	**F I**, 2, 4
6	$\quad \psi$	**RA**, 6–5
7	$\phi \to \psi$	$\to$I, 2–6

Examples

Exercise

Show that:

a) $\phi \to \psi \vdash \neg\phi \ \vee \ \psi$

b) $\neg(\phi \ \wedge \ \psi) \vdash \neg\phi \ \vee \ \neg\psi$

c) $\vdash \phi \to (\psi \to \phi)$

d) $\vdash (\phi \to (\psi \to \theta)) \to ((\phi \to \psi) \to (\phi \to \theta))$

e) $\vdash (\neg\psi \to \neg\phi) \to ((\neg\psi \to \phi) \to \psi)$

$\phi \to \psi \vdash \neg\phi \lor \psi$

1	$\phi \to \psi$	
2	$\quad \neg(\neg\phi \lor \psi)$	
3	$\quad\quad \neg\phi$	
4	$\quad\quad \neg\phi \lor \psi$	$\lor$I, 3
5	$\quad\quad$ **F**	**F** I, 2, 4
6	$\quad \neg\neg\phi$	$\neg$I, 3–5
7	$\quad \phi$	$\neg$E, 6
8	$\quad \psi$	$\to$E, 1, 7
9	$\quad \neg\phi \lor \psi$	$\lor$I, 8
10	$\quad$ **F**	**F** I, 2, 9
11	$\neg\neg(\neg\phi \lor \psi)$	$\neg$I, 2–10
12	$\neg\phi \lor \psi$	$\neg$E, 11

$\neg(\phi \wedge \psi) \vdash \neg\phi \vee \neg\psi$

1	$\neg(\phi \wedge \psi)$	
2	$\quad \neg(\neg\phi \vee \neg\psi)$	
3	$\quad\quad \neg\phi$	
4	$\quad\quad \neg\phi \vee \neg\psi$	$\vee$I, 3
5	$\quad\quad$ **F**	**F** I, 2, 4
6	$\quad \neg\neg\phi$	$\neg$I, 3–5
7	$\quad \phi$	$\neg$E, 6
8	$\quad\quad \neg\psi$	
9	$\quad\quad \neg\phi \vee \neg\psi$	$\vee$I, 3
10	$\quad\quad$ **F**	**F** I, 2, 9
11	$\quad \neg\neg\psi$	$\neg$I, 8–10
12	$\quad \psi$	$\neg$E, 11
13	$\quad \phi \wedge \psi$	$\wedge$I, 7, 12
14	$\quad$ **F**	**F** I, 1, 13
15	$\neg\neg(\neg\phi \vee \neg\psi)$	$\neg$I, 2, 14
16	$\neg\phi \vee \neg\psi$	$\neg$E, 15

$\vdash \phi \to (\psi \to \phi)$

1	ϕ	
2	$\quad\psi$	
3	$\quad\phi$	R, 1
4	$\psi \to \phi$	$\to$I, 2–3
5	$\phi \to (\psi \to \phi)$	$\to$I, 1–4

$\vdash (\phi \to (\psi \to \theta)) \to ((\phi \to \psi) \to (\phi \to \theta))$

1	$\phi \to (\psi \to \theta)$	
2	$(\phi \to \psi)$	
3	ϕ	
4	ψ	$\to$E, 2, 3
5	$\psi \to \theta$	$\to$E, 1, 3
6	θ	$\to$E, 4, 5
7	$\phi \to \theta$	$\to$I, 3–6
8	$(\phi \to \psi) \to (\phi \to \theta)$	$\to$I, 2–7
9	$\phi \to (\psi \to \theta) \to (\phi \to \psi) \to (\phi \to \theta)$	$\to$I, 1–8

$\vdash (\neg\psi \to \neg\phi) \to ((\neg\psi \to \phi) \to \psi)$

1	$\neg\psi \to \neg\phi$	
2	$\neg\psi \to \phi$	
3	$\neg\psi$	
4	ϕ	$\to$E, 2, 3
5	$\neg\phi$	$\to$E, 1, 3
6	**F**	**F I**, 4, 5
7	$\neg\neg\psi$	$\neg$I, 3–6
8	ψ	$\neg$E, 7
9	$(\neg\psi \to \phi) \to \psi$	$\to$I, 2–8
10	$(\neg\psi \to \neg\phi) \to (\neg\psi \to \phi) \to \psi$	$\to$I, 1–9